

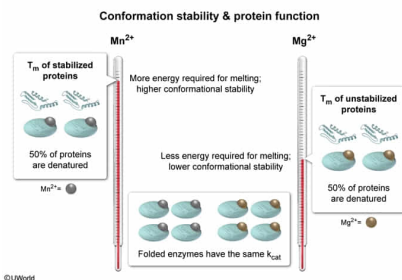
A certain enzyme uses Mn^{2+} as a cofactor under physiological conditions, and scientists hypothesize that Mn^{2+} is required for stabilization of tertiary structure but not for enzyme activity. Which of the following observations is consistent with this hypothesis?

- ✔ ☒ A. Substitution of Mg^{2+} in place of Mn^{2+} yields a decreased melting temperature and no change in the k_{cat} of folded proteins. [74%]
- ☐ B. Replacing Mn^{2+} with Cu^{2+} yields an unchanged melting temperature and a decreased k_{cat} of folded proteins. [7%]
- ☐ C. Using Ni^{2+} in place of Mn^{2+} yields a decreased melting temperature and a k_{cat} value of zero. [4%]
- ☐ D. Removing Mn^{2+} yields an increased melting temperature and no change in the k_{cat} of folded proteins. [13%]

Correct

75% answered correctly

Explanation:



The ability of a protein to maintain its three-dimensional folded form (tertiary structure) is known as **conformational stability**. Conformational stability can be measured by **melting temperature T_m** , the temperature at which half of the proteins in solution are folded and half are denatured. A **lower T_m** indicates *decreased* conformational stability because *less* thermal energy is required to **denature** the protein.

The ability of a folded enzyme to catalyze a reaction is measured by the **turnover number k_{cat}** , which represents the number of reactions catalyzed per second per active site in solution. A change in the conformational stability of an enzyme **does not necessarily affect function**, as long as the ambient temperature is low enough that the protein can fold properly.

In this scenario, scientists hypothesized that Mn^{2+} is required for stabilization of tertiary structure (ie, for conformational stability), but *not* for enzyme activity. If this hypothesis is correct, then when Mn^{2+} is removed or replaced by a different ion, the T_m of the protein will decrease due to decreased conformational stability. However, the k_{cat} will not change because the ability of folded proteins to catalyze the reaction has not changed. Therefore, substitution of a different metal ion such as Mg^{2+} should **decrease T_m without affecting k_{cat}** .

(Choice B) If Cu^{2+} yielded the same T_m as Mn^{2+} , it would indicate that Mn^{2+} is *not* required for conformational stability. A decreased k_{cat} would indicate that Mn^{2+} is important for catalysis.

(Choice C) A decreased T_m indicates decreased conformational stability, but a k_{cat} of zero when Ni^{2+} replaces Mn^{2+} would indicate that Mn^{2+} is required for catalysis.

(Choice D) An increased T_m in the absence of Mn^{2+} would indicate that Mn^{2+} is *not* required for conformational stability and actually *reduces* stability.

Educational objective:

The conformational stability of a protein is its ability to maintain its three-dimensional folded form and can be measured by the melting temperature (T_m) at which half of the proteins in solution are folded and half are denatured. Conformational stability does not necessarily affect protein function, as long as the protein is maintained at a temperature that allows proper folding.

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